

SAPNIL PATEL

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EDUCATION

University of Southern California (USC)

January 2025 – December 2026

Master of Science in Applied Data Science

Coursework: Data Management, Machine Learning for Data Science, Predictive Analytics, Data Mining

CHARUSAT University

October 2020 – April 2024

Bachelor of Technology in Computer Engineering

Coursework: Artificial Intelligence, Natural Language Processing, Data Structures and Algorithms, Networking, Big Data Analytics, Computer Vision, Database Systems, Linear Algebra

WORK EXPERIENCE

Pacific Production Services, Inc.

Los Angeles, USA

AI Automation Engineer

June 2026 – Present

- Architected concurrent, high-throughput backend microservices using Python and FastAPI to automate document parsing, reducing processing latency and saving **\$25,000+** annually
- Automated **100%** of rental-agreement generation from live scheduling data (Python, OpenXML) via an idempotent, **unit-tested** pipeline with content-hash change detection for replay-safe regeneration saving **\$55000+** annually

USC-CeNEC Lab

Los Angeles, USA

AI Research Assistant

March 2025 – Present

- Engineered distributed evaluation engines to benchmark LLM outputs across **1,050+** experiments, developing optimized data-validation scripts that improved throughput by **50%**
- Implemented custom inference pipelines with **Ollama** to integrate fine-tuned local models into automated agent frameworks, minimizing network call overhead and optimizing memory footprint

Nivaan Infotech

Visnagar, India

Software Developer

January 2024 – October 2024

- Developed conversational backends over a **PostgreSQL** order-processing service, handling asynchronous user events to eliminate **1-2 hours** of manual request handling daily
- Deployed **predictive modeling** and **PPO-based reinforcement learning** for user personalization, lifting engagement **25%** under differential privacy via **TensorFlow Privacy**

PROJECTS

Minisearch | PostgreSQL, REST API, Prometheus

[GitHub](#)

- Designed a search engine serving BM25-ranked queries at **1-13 ms p50 latency** over 20K documents via a positional inverted index with sorted posting-list intersection, phrase matching, and bounded min-heap top-K selection
- Sustained **7,400 pages/sec** crawl throughput at **40 MB peak memory** by designing a parallel async fetch pipeline with bounded worker pools, backpressure, a priority frontier enforcing per-host politeness, and graceful shutdown
- Cut URL-deduplication memory **99x** versus a hash set (117 KiB per 100K URLs) by implementing a bloom filter from scratch and validating its measured 1.02% false-positive rate within 2% of theory

LLM Inference Optimizer | GCP

[GitHub](#)

- Engineered a distributed LLM inference service (**LLaMA-3/Mistral, PyTorch**) with dynamic request batching, KV-cache reuse, and FP16/BF16 mixed precision for **2.3x** throughput and **38%** lower GPU memory over FP32
- Implemented tensor-parallel serving across GPUs via DeepSpeed with an asynchronous batching scheduler and admission control for backpressure, sustaining low tail latency under high concurrency
- Built an observability and autoscaling stack (PyTorch Profiler, Prometheus, Grafana) deployed on GCP via Google Kubernetes Engine (GKE) with horizontal pod autoscaling for fault-tolerant, zero-downtime serving

Silicon Validation Log Analysis Agent | LangGraph, Pinecone

[GitHub](#)

- Built a **multi-agent** AI debug assistant (**LangGraph, Pinecone RAG**) that automated the post-silicon failure-triage workflow across **10M+** validation logs, cutting engineering debugging time **40%**
- Applied clustering, anomaly detection, and **LLM** classification to identify **85%+** recurring failure patterns

TECHNICAL SKILLS

- **Languages:** Python, C++, Java, JavaScript, C, SQL, Bash, Kotlin, Scala
- **Backend & Systems:** Flask, REST APIs, Docker, Kubernetes (GKE), CI/CD, Git, Linux, Prometheus, Grafana, pytest, CUDA
- **Cloud & Databases:** AWS (SageMaker, EC2, S3), GCP, PostgreSQL, MongoDB, Redis, Pinecone
- **ML & Deep Learning:** PyTorch, TensorFlow, Keras, Scikit-learn, Transformers, HuggingFace, XGBoost, LightGBM, OpenCV, NLTK, NumPy, Pandas, DeepSpeed
- **LLM & Agents:** LangChain, LangGraph, CrewAI, Ollama, RAG, vector databases, LLM fine-tuning, Claude Code